E-Commerce Data Warehouse in MySQL

1. About the project

In this project I took the order data of Olist, an online marketplace in Brazil, and rebuilt it as a small data warehouse in MySQL. Then I used the warehouse to answer six business questions with SQL.

The data is the Brazilian E-Commerce Public Dataset from Kaggle. It has about 100,000 orders from September 2016 to August 2018, spread over several files. I worked on my own laptop with MySQL 8 and MySQL Workbench. It is a learning project on public data, not a live company system.

2. Problem statement

The Olist data comes in separate files. The order items file has the price and the product, but it does not have the customer or the date. The customer is in one file and the date is in another. The category names are in Portuguese.

Because of this, even a simple question such as "how much did we sell in March?" needs several joins every time, and nothing checks that the joins are correct.

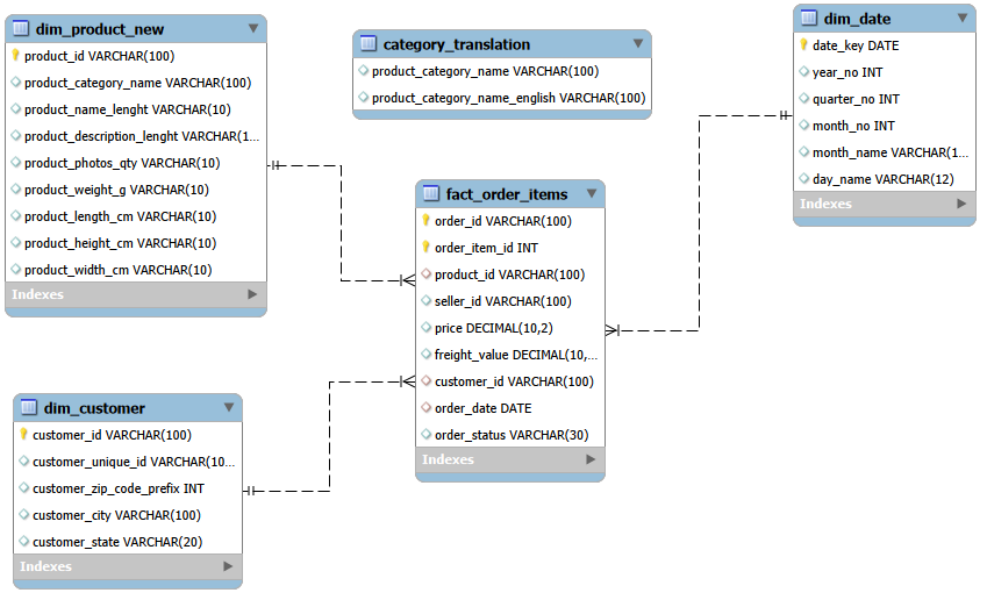
So the problem was: put the data in one clean structure, fill it correctly, check it, and then answer these questions.

1. Which product categories earn the most?
2. How do orders and revenue change month by month?
3. Which states buy the most?
4. Which sellers earn the most?
5. In which categories is shipping expensive compared to the price?
6. How many orders are delivered, cancelled or still in progress?

3. What I did

I designed a star schema. The fact table is in the middle and has one row for every item sold. The tables around it describe the product, the customer and the date.

Table	Role	What it holds
fact_order_items	Fact	One row per item sold: price, freight, product, seller, customer, date, status
dim_product_new	Dimension	Product and its category name
dim_customer	Dimension	Customer city and state
dim_date	Dimension	One row per day, 852 days: year, quarter, month, day name
category_translation	Lookup	Portuguese to English category names



4. How I did it

I worked in the following order.

Step	What I did	Result
1. Load	Imported the CSV files with the Import Wizard and made a backup of the fact table	Raw tables and a backup
2. Staging	Imported the orders file as plain text into a staging table, so the import cannot fail	99,441 rows
3. Join	Indexed order_id, then one UPDATE copied customer, date and status into the fact table	112,650 rows in about 26 seconds
4. Date table	Built it with a recursive query, one row for every day from 1 Sep 2016 to 31 Dec 2018	852 rows
5. Translation	Loaded the Portuguese to English category names	About 71 rows
6. Foreign keys	Added keys from the fact table to product, customer and date	All 3 accepted
7. Validation	Counted fact rows with no matching product, customer or date, and rows with missing values	All counts were 0
8. Queries	Wrote the six queries	0.2 to 1.1 seconds each

The staging table was needed because the customer and the date were in the orders file, not in the order items file. I loaded the orders file as text first so that a strange value could not stop the import, and then joined it to the fact table on the order ID.

I added the foreign keys last. MySQL checks every row when a key is created, so it would have refused if any row had no match. The validation query then confirmed the same thing: no missing products, customers or dates.

Revenue means the sum of the item price for orders with the status "delivered". Freight is not included, and cancelled or unfinished orders are not counted. All amounts are in Brazilian reais

(R$).

5. Results

The warehouse has 96,478 delivered orders and 110,197 delivered items, from 15 September 2016 to 29 August 2018. Together they brought in R$ 13,221,498 in revenue. The average order value is R$ 137.04.

Categories. These are the five categories with the highest revenue. The top five together are about 40% of the revenue, and the top ten about 62%. bed_bath_table sells the most items but is only third in revenue.

Category	Items sold	Revenue (R$)
health_beauty	9,465	1,233,132
watches_gifts	5,859	1,166,177
bed_bath_table	10,953	1,023,435
sports_leisure	8,431	954,853
computers_accessories	7,644	888,725

States. São Paulo is far ahead with R$ 5,067,633 from 40,501 orders, about 38% of all revenue. Rio de Janeiro (R$ 1,759,651) and Minas Gerais (R$ 1,552,482) come next. These three states together make up about 63%.

Order status. Almost every order was delivered. Only 461 orders (0.5%) were cancelled.

Status	Orders	Share
delivered	96,478	97.8%
shipped	1,106	1.1%
canceled	461	0.5%
invoiced	312	0.3%
processing	301	0.3%
unavailable	6	0.0%
approved	2	0.0%

Freight. Among categories with at least 100 items sold, shipping is the biggest share of the price for christmas_supplies (36.5%), signaling_and_security (30.4%) and electronics (29.5%). These are mostly low-priced items, so a fixed shipping cost looks large next to the price.

Sellers. The top seller earned R$ 226,988 from 1,148 items. The second seller earned almost the same, R$ 217,940, from only 400 items, so that seller sells more expensive things.

Months. The data starts thin: 1 order in September 2016, 265 in October and 1 in December. In 2017 the monthly orders rise from 750 in January to 2,546 in March. I do not read any trend into 2016.

These are observations from one dataset. I did not test why they happen.

6. What is missing and what I would do next

There are some gaps in this project. Each one has a next step.

Gap	Next step
No seller table, so sellers are only an ID	Add a seller dimension from the Olist sellers file
An older product table (dim_product) is not used	Remove it from the database
Product weight and size are stored as text	Convert them to numbers and compare freight with weight
Revenue ignores returns and discounts	Check the totals against the payments file
Only five of the Olist files are used	Add the reviews file and compare ratings with delivery time
Tested on about 112,000 rows only	Test on a larger dataset
No dashboard	Connect the warehouse to Power BI and build one for the same six questions